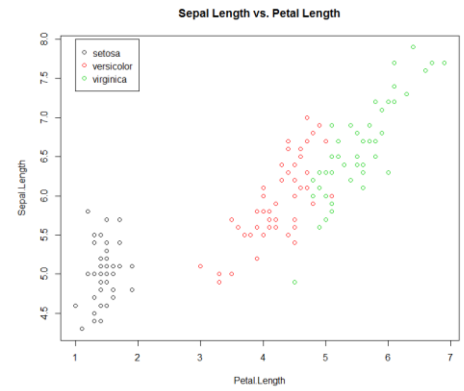
Introduction to ggplot2

http://ggplot2.tidyverse.org/reference/index.html

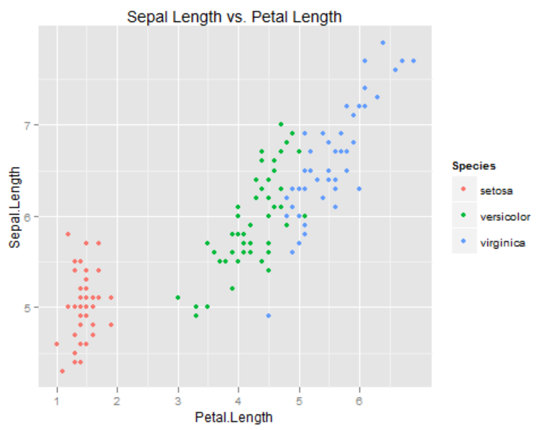
```
BiocManager::install("ggplot2")
```

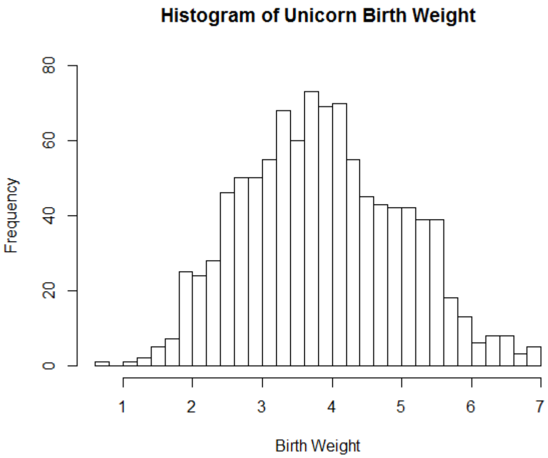
Overview

- Base graphics
  - Graphical tools already included in R
  - Simple, fast, exploratory graphics
  - Important to know

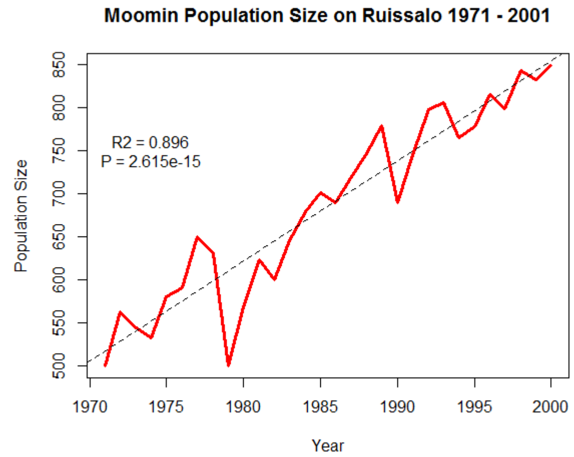


- `ggplot2`
  - More complex, higher quality graphics
  - Fashionable to know
  - Easier to master?

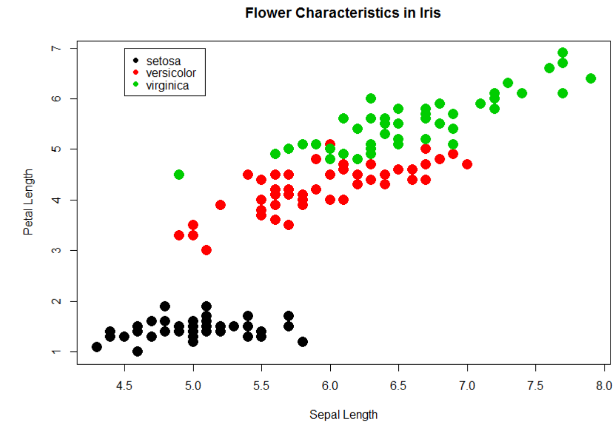




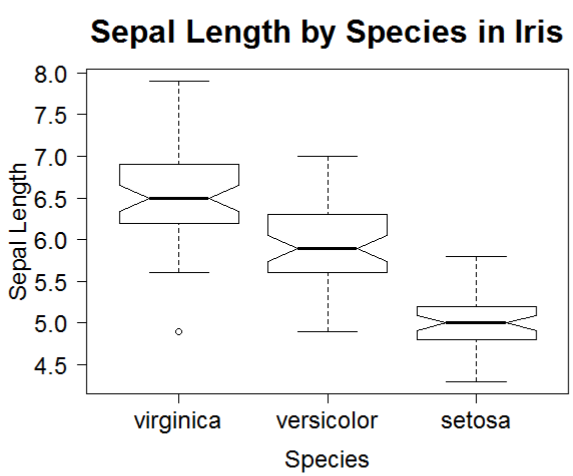
1. Basic Histogram



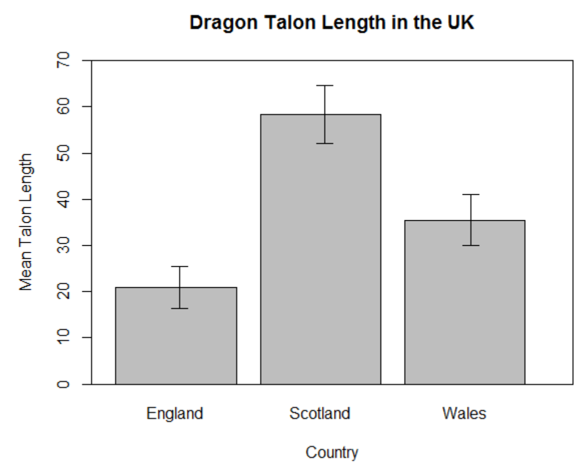
2. Line Graph with Regression



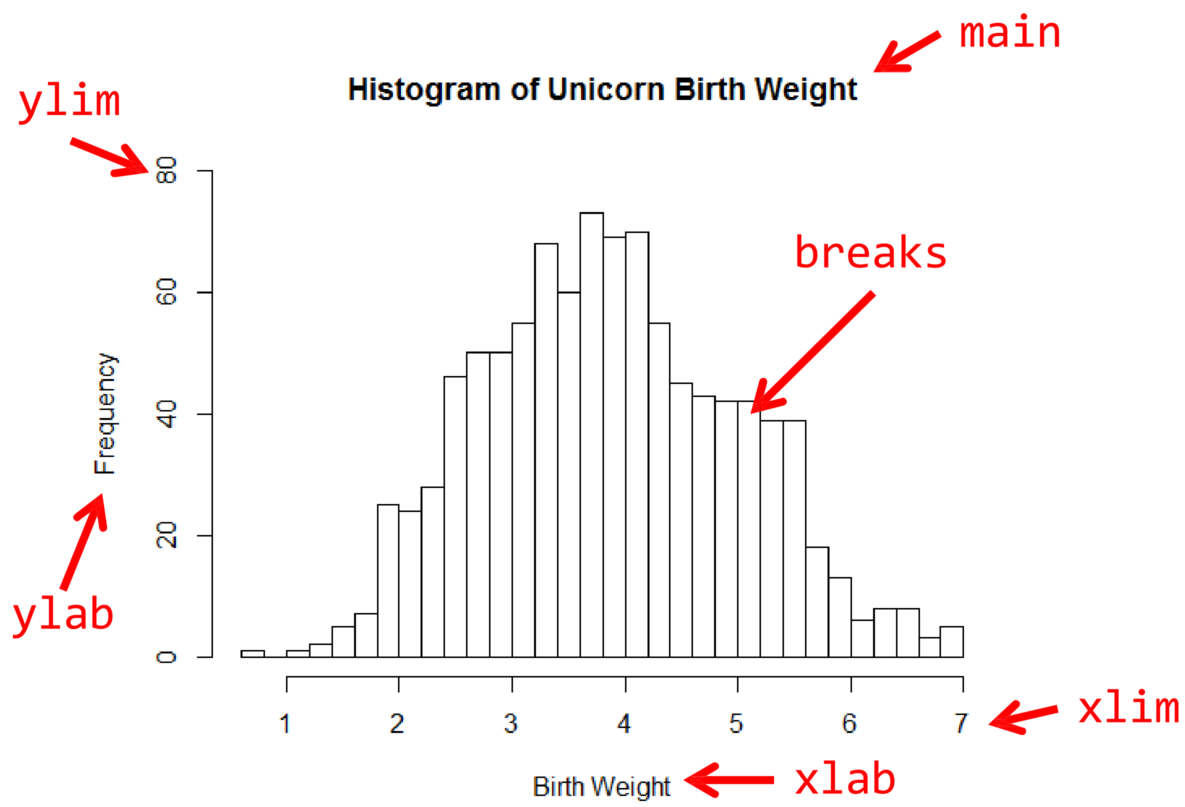
3. Scatterplot with Legend



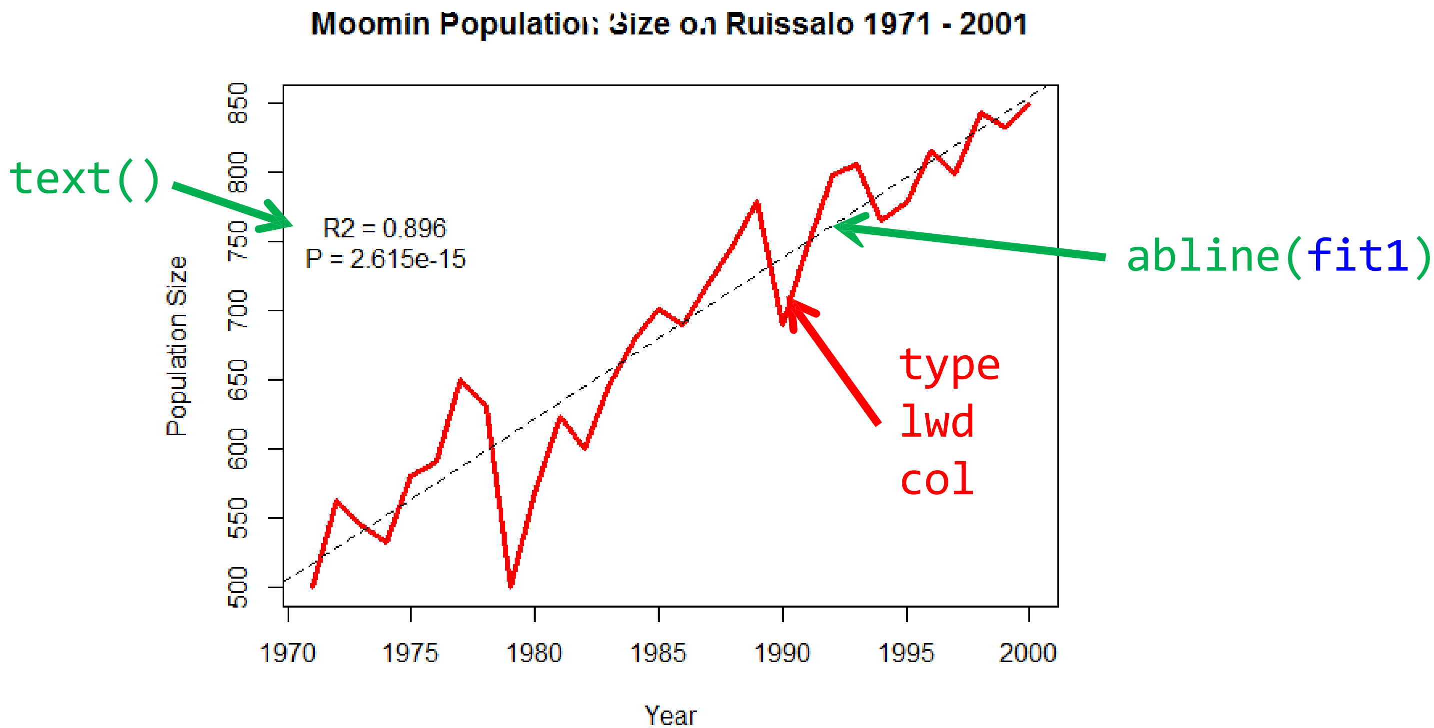
4. Boxplot with reordered/ formatted axes



5. Barplot with Error Bars

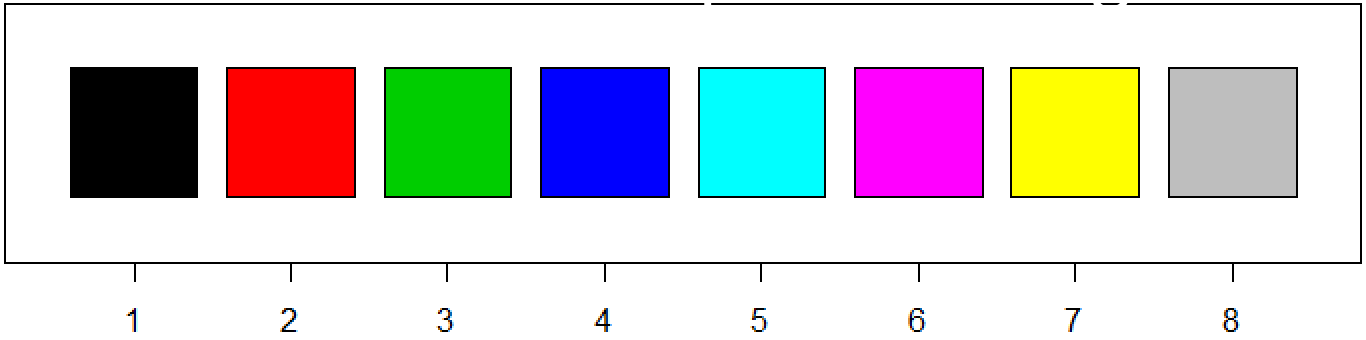



```
#~~ FINAL PLOT:

hist(unicorns$birthweight,                          # x value
     breaks = 40,                                   # number of cells
     xlab = "Birth Weight",                         # x-axis label
     main = "Histogram of Unicorn Birth Weight",    # plot title
     ylim = c(0,80))                                # limits of the y axis (min,max)

```

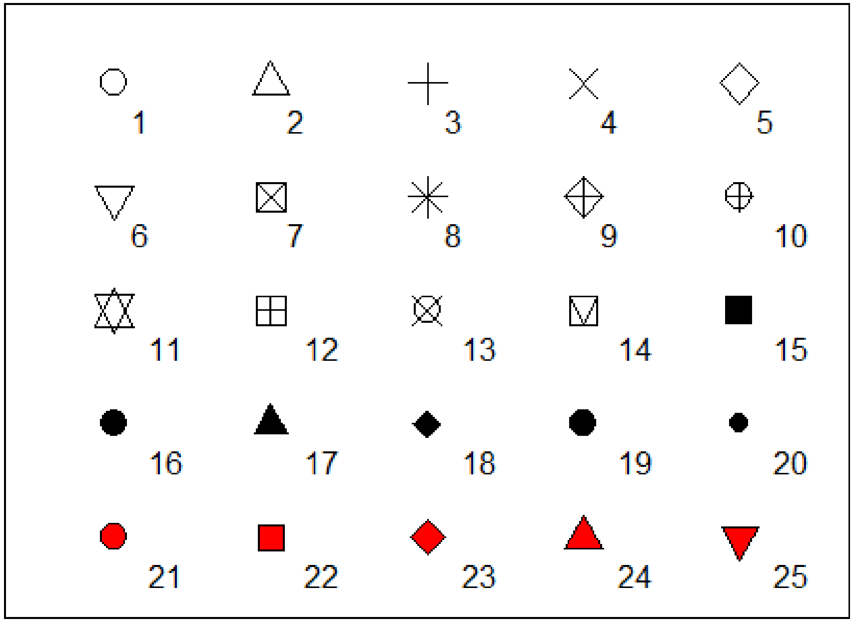


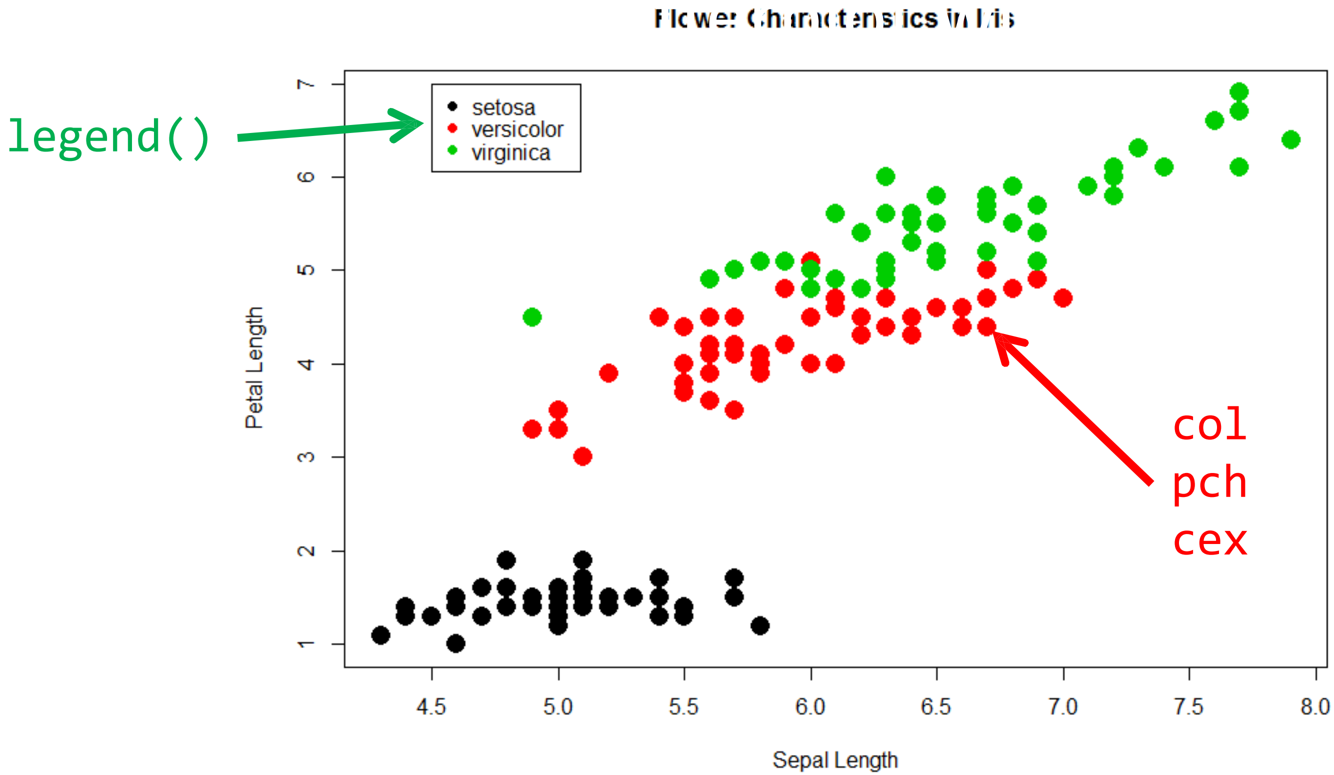

```
#~~ FINAL PLOT Script

plot(moomins$Year, moomins$PopSize,                               # x variable, y variable
     type = "l",                                                  # draw a line graphs
     col = "red",                                                 # red line colour
     lwd = 3,                                                     # line width of 3
     xlab = "Year",                                               # x axis label
     ylab = "Population Size",                                    # y axis label
     main = "Moomin Population Size on Ruissalo 1971 - 2001")     # plot title
fit1 <- lm (PopSize ~ Year, data = moomins)              # carry out a linear regression
abline(fit1, lty = "dashed")                             # add the regression line to the plot
text(x=1974,y=750,labels="R2 = 0.896\nP = 2.615e-15")    # add a label to the plot at coordinates (x,y)

```

col

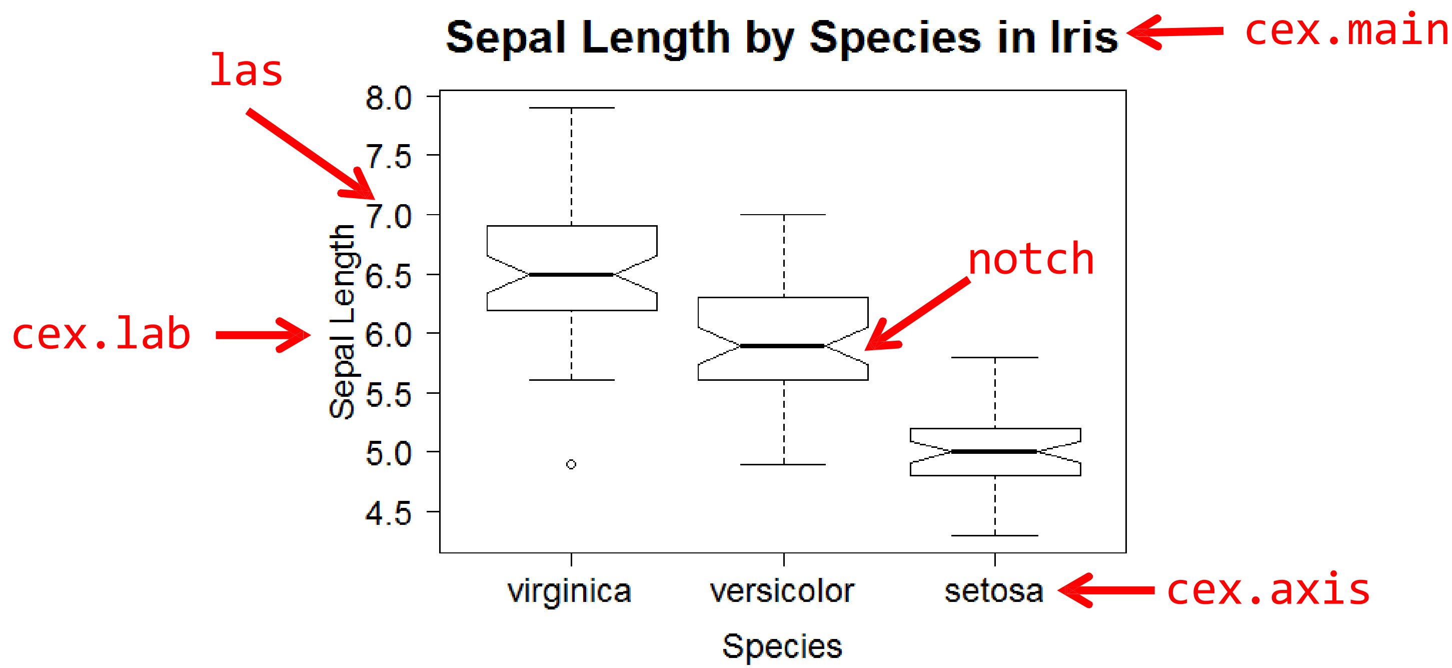


pch



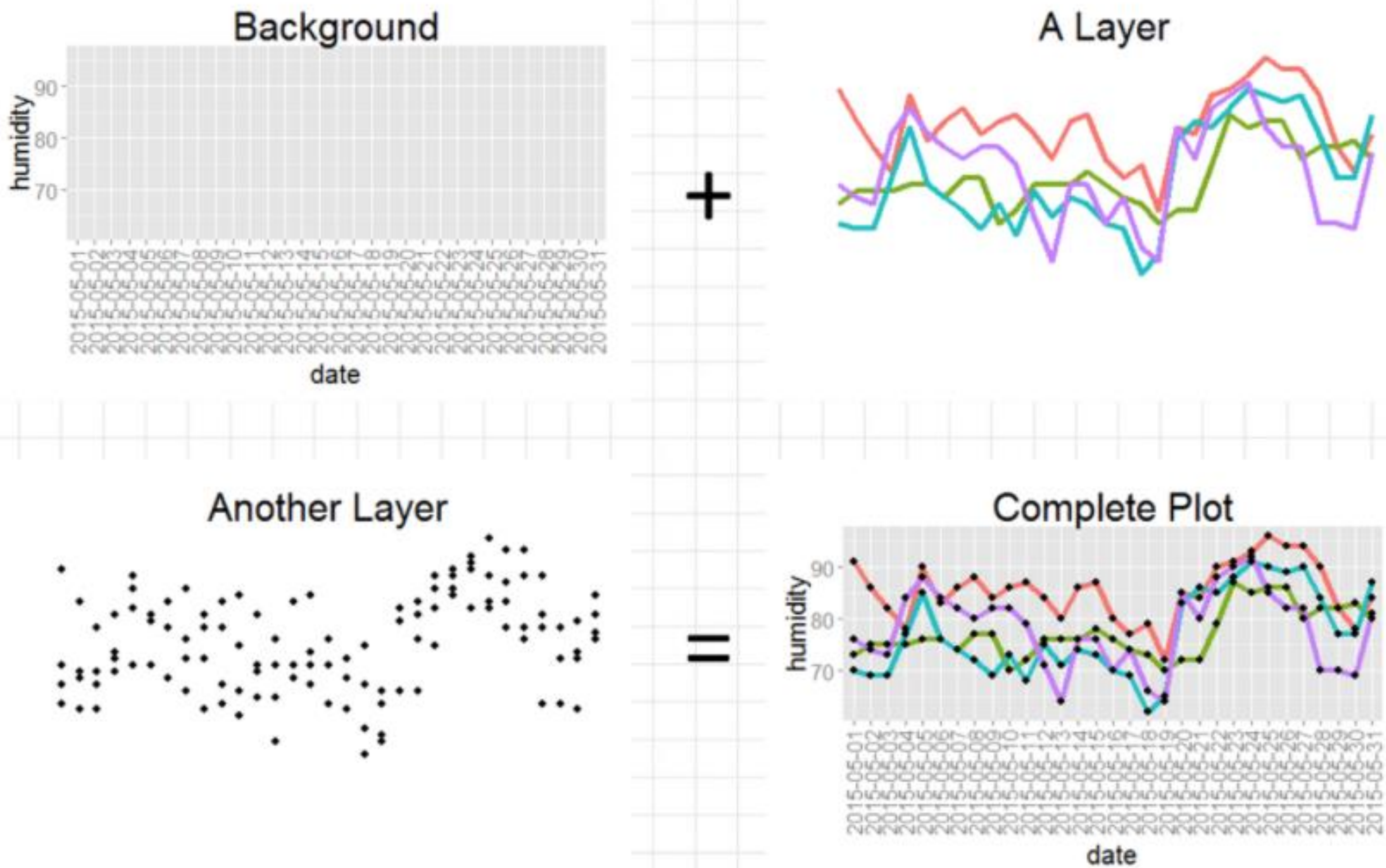



```
# FINAL PLOT

plot(iris$Sepal.Length, iris$Petal.Length,          # x variable, y variable
     col = iris$Species,                            # colour by species
     pch = 16,                                      # type of point to use
     cex = 2,                                       # size of point to use
     xlab = "Sepal Length",                         # x axis label
     ylab = "Petal Length",                         # y axis label
     main = "Flower Characteristics in Iris")       # plot title

legend (x = 4.5, y = 7, legend = levels(iris$Species), col = c(1:3), pch = 16)
# legend with titles of iris$Species and colours 1 to 3, point type pch at coords (x,y)
```




```
#~~ FINAL PLOT

iris$Species<-factor(iris$Species, levels = c("virginica","versicolor","setosa"))

boxplot(iris$Sepal.Length ~ iris$Species,                   # x variable, y variable
        notch = T,                                          # Draw notch
        las = 1,                                            # Orientate the axis tick labels
        xlab = "Species",                                   # X-axis label
        ylab = "Sepal Length",                              # Y-axis label
        main = "Sepal Length by Species in Iris",           # Plot title
        cex.lab = 1.5,                                      # Size of axis labels
        cex.axis = 1.5,                                     # Size of the tick mark labels
        cex.main = 2)                                       # Size of the plot title
```

The Anatomy of a Plot



A Straightforward Way to Create Visualizations

- Grammar of Graphics provides a framework to streamline the description and creation of graphics
- For a given dataset to be displayed:
  - Map variables to aesthetics
  - Define Layers
    - A representation (a ‘geom’) ie line, boxplot, histogram,…
    - Associated statistical transformation ie counts, model,…
  - Define Scales
    - Color, Shape, axes,…
  - Define Coordinates
  - Define Facets

ggplot2簡介

- 當前最多人使用的視覺化R套件 (2015年，最受歡迎的R套件之一)
- R環境下的繪圖套件
- 取自 “The Grammar of Graphics” (Leland Wilkinson, 2005)
- 設計理念
  - 採用圖層系統
  - 用抽象的概念來控制圖形，避免細節繁瑣
  - 圖形美觀

ggplot2簡介

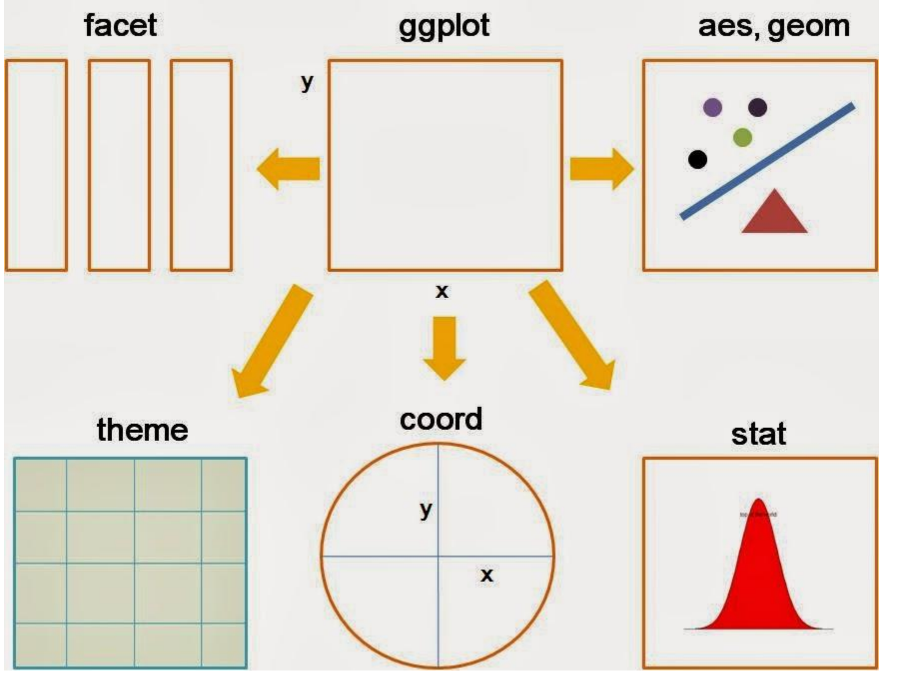
- ggplot 是以繪圖文法概念為基礎所發展出來的一套 R 繪圖系統，可繪製各種高品質圖形的 R 繪圖套件，是一套相當受歡迎的繪圖工具。
- 在使用上，ggplot 這種繪圖系統不同於以往傳統式的 R 繪圖方式，使用者不會受限於既有的固定功能，可以依照自己的需求組合各種圖形組件、任意調整圖形，產生各種變化，相當富有彈性，而其畫出來的圖形品質也非常好，通常使用預設的設定值就可以很容易地產生出版品水準的圖形。

ggplot2簡介

- 在傳統的 R 繪圖系統之下，一般的繪圖工具所提供的只是一些固定的繪圖功能，頂多加上一些微調參數讓使用者做些微的調整，若要自行設計新的圖形，使用者能使用的繪圖組件只有資料點、線條等低階的元素，既有的高階圖形完全無法重複使用。
- 在 ggplot 系統之下，繪圖文法的設計可以讓使用者以更高階的思維模式來設計繪圖，有許多既有的高階圖形元素可以使用，例如資料的呈現以及統計上的資料轉換方式等，使用者只要運用這些 ggplot 系統下的元件，就可以快速組裝出適用於自己的圖形。

ggplot2基本架構

- 資料 (data) 和映射 (mapping)
- 幾何對象 (geometric)
- 座標尺度 (scale)
- 統計轉換 (statistics)
- 座標系統 (coordinate)
- 圖層 (layer)
- 刻面 (facet)
- 主題 (theme)



R的ggplot2基本概念

-要將資料畫到螢幕上，首先要有資料；再來是指定資料要怎樣繪圖；最後才是調整圖片 。

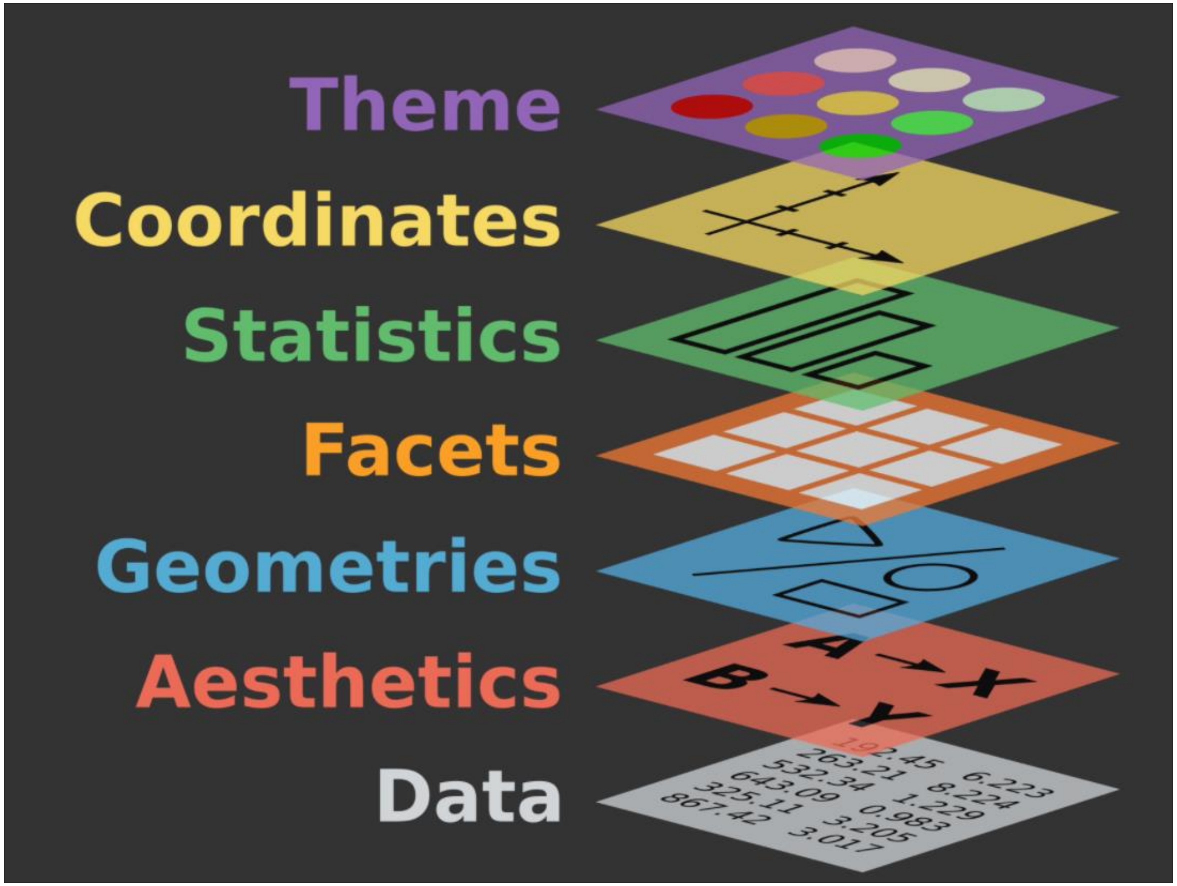
R的ggplot2這個繪圖套件根據下列幾個概念，將資料描繪到圖上：

- 資料來源（Data）：指定資料來源
- 美學對應（Aesthetics）：選定要繪圖的資料點。例如說xy散佈圖，就要指定哪些資料點是x軸、哪些點是要放y軸
- 幾何圖案（Geometries）：要畫成哪種圖，像是直方圖、散佈圖、箱型圖、柱狀圖、折線圖等等。更多種繪圖方式可以參考ggplot2的Geoms列表
- 數值範圍（Scales）：根據數值範圍可以上不同的顏色，例如說數值由小而大從藍到紅描繪在圖上。參考ggplot2的Scales列表
- 繪圖面（Facets）：將很多張圖放在一起比較，參考ggplot2的Faceting列表
- 統計轉換（Statistics）：將指定的資料轉換成各種統計量，像將連續轉成離散
- 座標系統（Coordinates）：一般是直角座標系統，也可以對調座標，或用極座標系統、甚至是地圖的座標系統。參考ggplot2的Coordinate systems。
- 繪圖主題（Theme）：資料以外的繪圖物件，像是說明文字的調整等等

簡單的圖只要前面三個Data + Aesthetics + Geometries就可以畫出來了。

ggplot 繪圖文法

- 繪圖文法（grammar of graphics）的概念最早是由 Wilkinson 所提出來的，其主要用來解釋統計圖形（statistical graphic）的概念，而後來 Wickham 則是以 Wilkinson 的繪圖文法理論為基礎，做了一些調整後應用於 R 語言中，發展出了 ggplot 這個繪圖系統。
- ggplot 系統將圖形視為一個從資料開始的一連串轉換，轉換的過程中間有很多道程序，包含幾何圖案、顏色與大小等美學屬性、統計量的計算與座標系統等，一個圖形就是原始資料結合這些轉換之後的結果。



ggplot2 基本語法

```
ggplot(data=..., aes(x=..., y=...)) + geom_xxx(...) +
stat_xxx(...) + facet_xxx(...) + theme(...)
```

- ggplot 描述 data 從哪來
- aes 描述圖上的元素跟 data 之類的對應關係
- geom_xxx 描述要畫圖的類型及相關調整的參數
- 常用的類型諸如：
  geom_bar, geom_points, geom_line, geom_polygon

注意事項

- 使用 data.frame 儲存資料 (不可以丟 matrix 物件)
- 使用 long format (利用reshape2/tidyr將資料轉換成 1 row = 1 observation)
- 文字型態的資料預設依 ascii 編碼順序做排序

寬資料

	ClassA	ClassB	ClassC
1	87	83	80
2	64	97	86
3	89	97	80
4	82	99	80
5	59	92	84

(a) `grades1` (unstacked form)

長資料

	values	ind
1	87	ClassA
2	64	ClassA
3	89	ClassA
4	82	ClassA
5	59	ClassA
6	83	ClassB
7	97	ClassB
8	97	ClassB
9	99	ClassB
10	92	ClassB
11	80	ClassC
12	86	ClassC
13	80	ClassC
14	80	ClassC
15	84	ClassC

(b) `grades2` (stacked form)

Layers

- Add layers with “+”
- Layers include any of the grammatical elements above (geoms, stats, facets, etc)
- We can store ggplot2 graph objects in R objects to keep code compact

Various functions

```
library(ggplot2)
# list all geom
ls(pattern = '^geom_', env = as.environment('package:ggplot2'))
```

```
 [1] "geom_abline"     "geom_area"       "geom_bar"
 [4] "geom_bin2d"      "geom_blank"      "geom_boxplot"
 [7] "geom_contour"    "geom_crossbar"   "geom_density"
[10] "geom_density2d"  "geom_dotplot"    "geom_errorbar"
[13] "geom_errorbarh"  "geom_freqpoly"   "geom_hex"
[16] "geom_histogram"  "geom_hline"      "geom_jitter"
[19] "geom_line"       "geom_linerange"  "geom_map"
[22] "geom_path"       "geom_point"      "geom_pointrange"
[25] "geom_polygon"    "geom_quantile"   "geom_raster"
[28] "geom_rect"       "geom_ribbon"     "geom_rug"
[31] "geom_segment"    "geom_smooth"     "geom_step"
[34] "geom_text"       "geom_tile"       "geom_violin"
[37] "geom_vline"
```

Aesthetics

- Variables are **mapped** to **aesthetics**
  - Variables are translated to visual elements of the graph
- The `aes` function maps variables to aesthetics
  - `aes` can be specified inside many `ggplot2` functions (geoms, stats, etc.)
- Commonly used aesthetics
  - `x` - position on x-axis
  - `y` - position on y-axis
  - `color` – either color of object itself or color of outline (see fill)
  - `fill` – fill color of object
  - `alpha` – transparency
  - `shape` – marker shape
  - `size` – size of objects (radius)
  - `linetype` – how lines are drawn

ggplot2 uses three components to construct a graph.

1. **Layers**: data with aesthetic properties
   - Data: iris
   - Aesthetic properties of the data

     `x = Sepal.Length, y = Petal.Length, colour = Species`

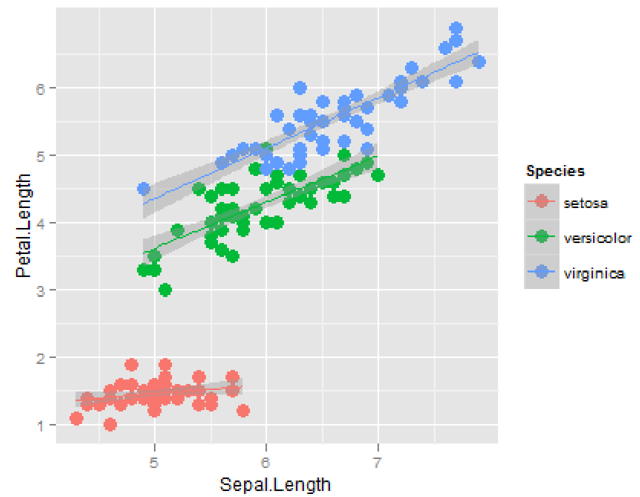
2. **Geoms**: the type of plot you make.
   - a line graph, a scatterplot, a boxplot

3. **Stats:** statistical transformations
   - e.g. assigning data to bins, smoothing lines, etc.
   - Every geom has a default statistic, so this is not always specified.

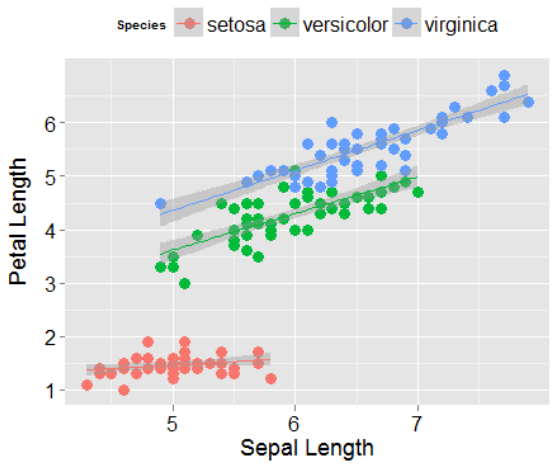
- This should become clearer as we go along.

Overview

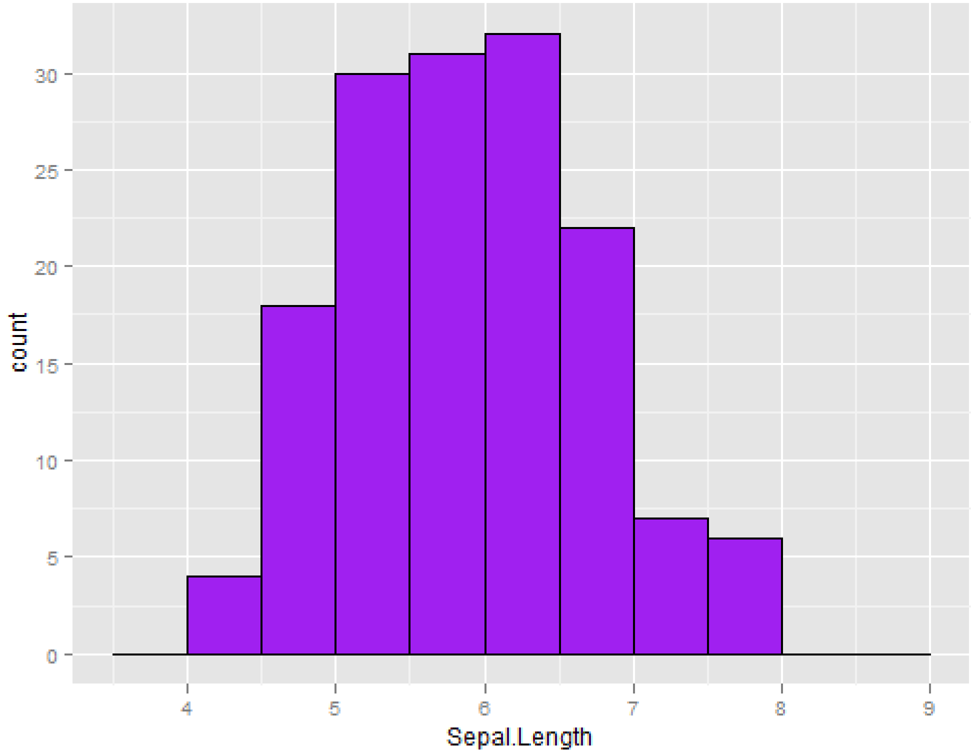
1. **Learn how to build plots**
   - **Data + aesthetic values**
   - **Geoms**
   - **Stats**
   - **Faceting and Dodging**



2. **Customise the layout**



1. Histogram

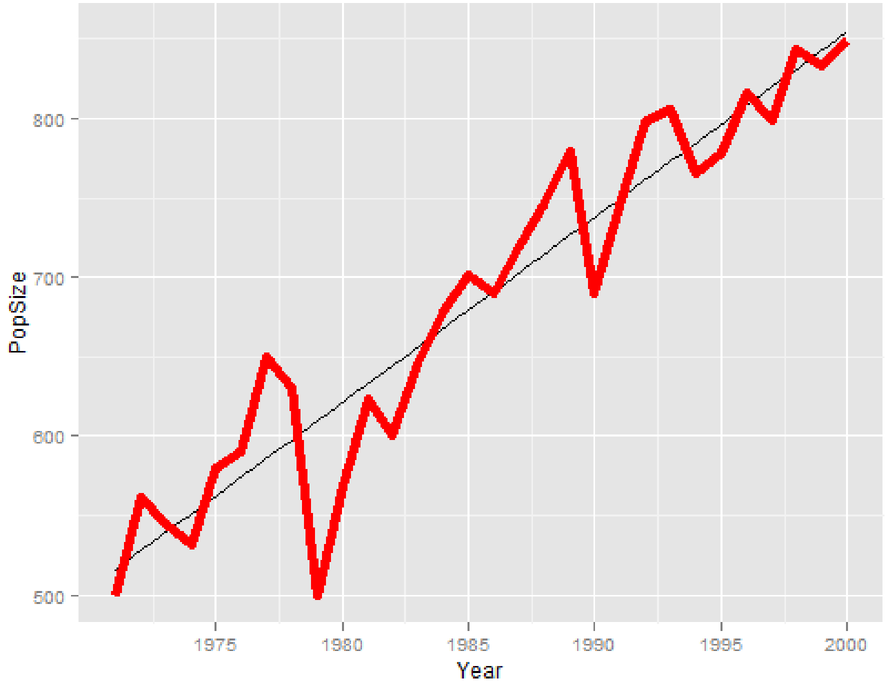



```
ggplot(iris, aes(x = Sepal.Length)) +
geom_histogram(binwidth = 0.5, col = "black", fill = "purple")
```

← DATA

← GEOM

2. Line Graph



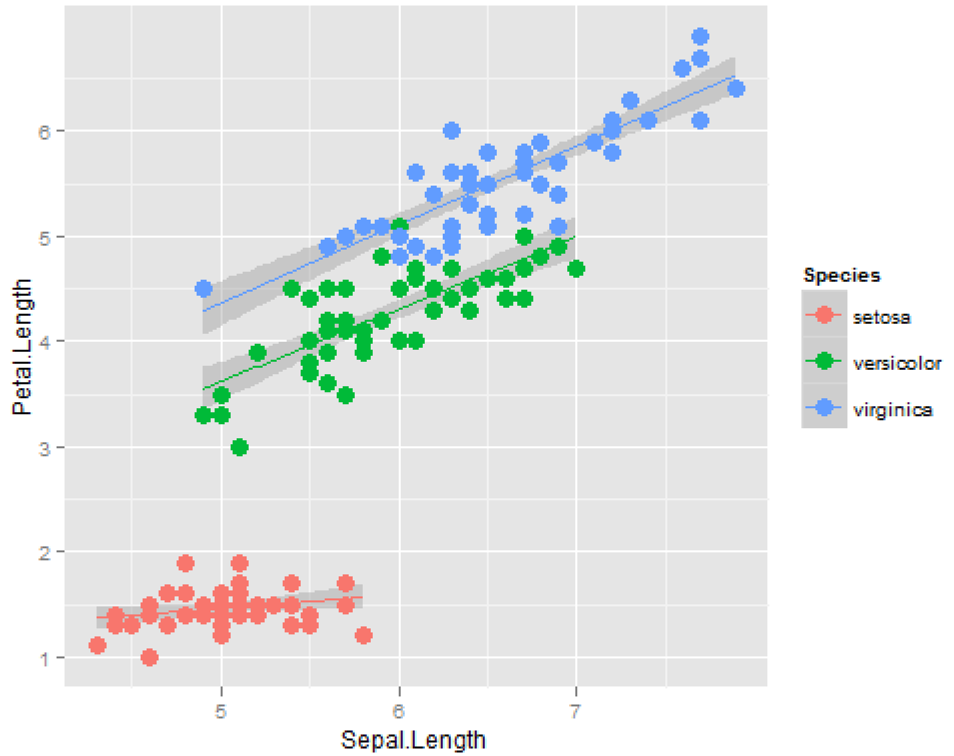

```
ggplot(moomins, aes(x = Year, y = PopSize)) +
  stat_smooth(method = "lm", col = "black", se = F) +
  geom_line(col = "red", size = 2)
```

DATA

STAT

GEOM

3. Scatterplot



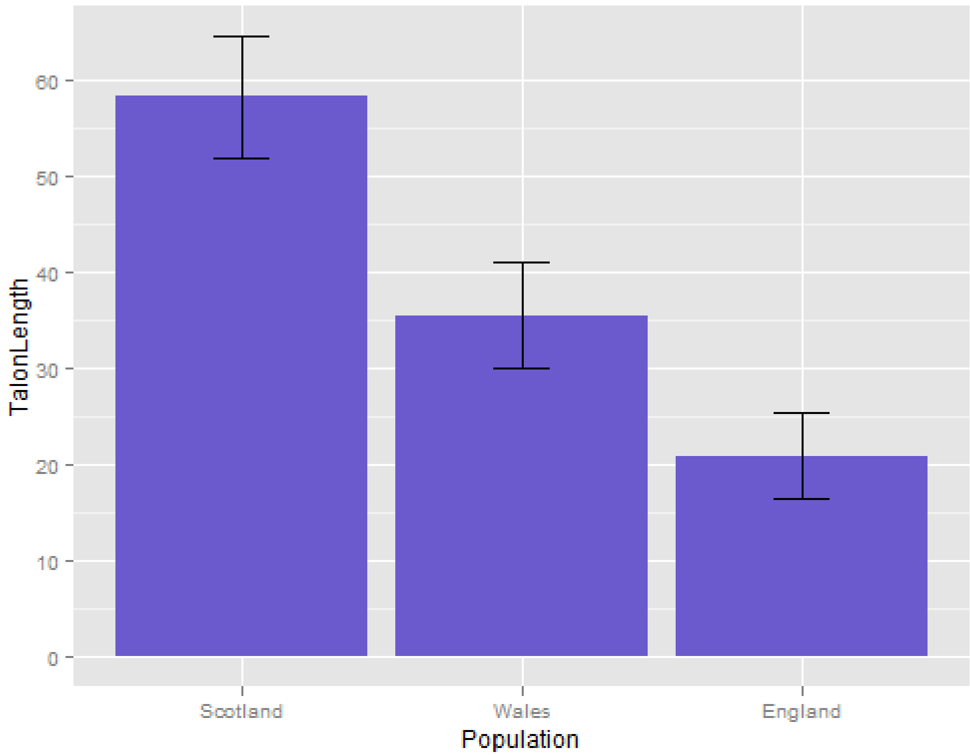
DATA

```
ggplot(iris, aes(x=Sepal.Length, y=Petal.Length, col=Species)) +
  stat_smooth(method = "lm") +
  geom_point(size = 4)
```

STAT

GEOM

4. Barplot



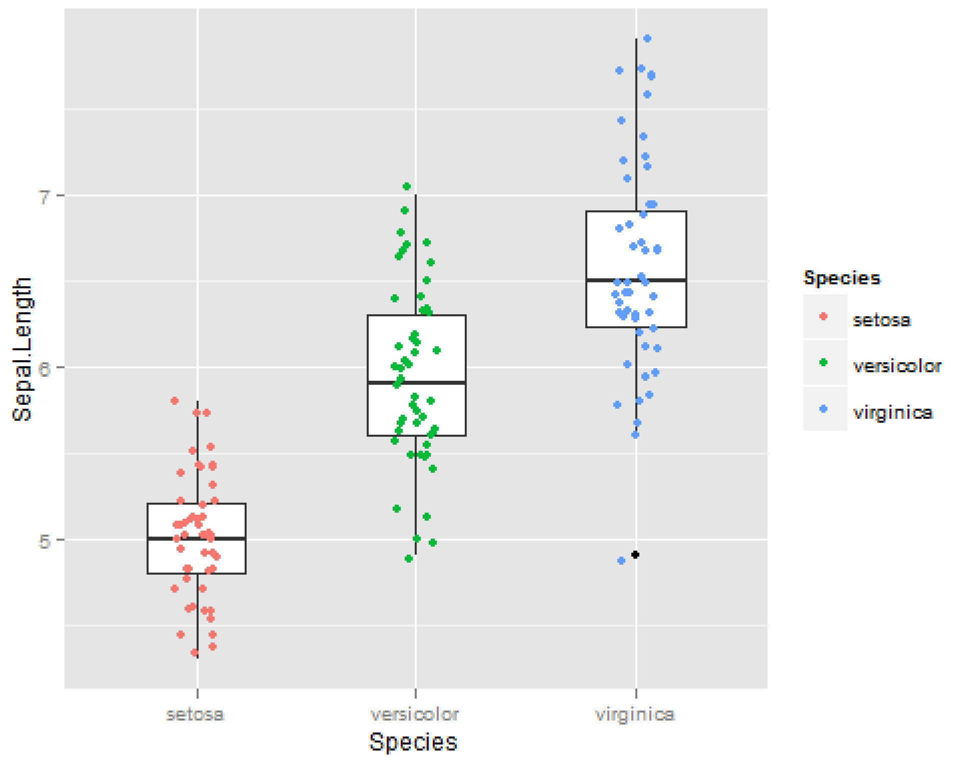

```
ggplot(dragons, aes(x = Population, y = TalonLength))
  + geom_bar(fill = "slateblue") +
  geom_errorbar(aes(ymax = TalonLength + SE,
       ymin = TalonLength - SE), width = 0.2)
```

DATA

GEOM

GEOM

4. Boxplot



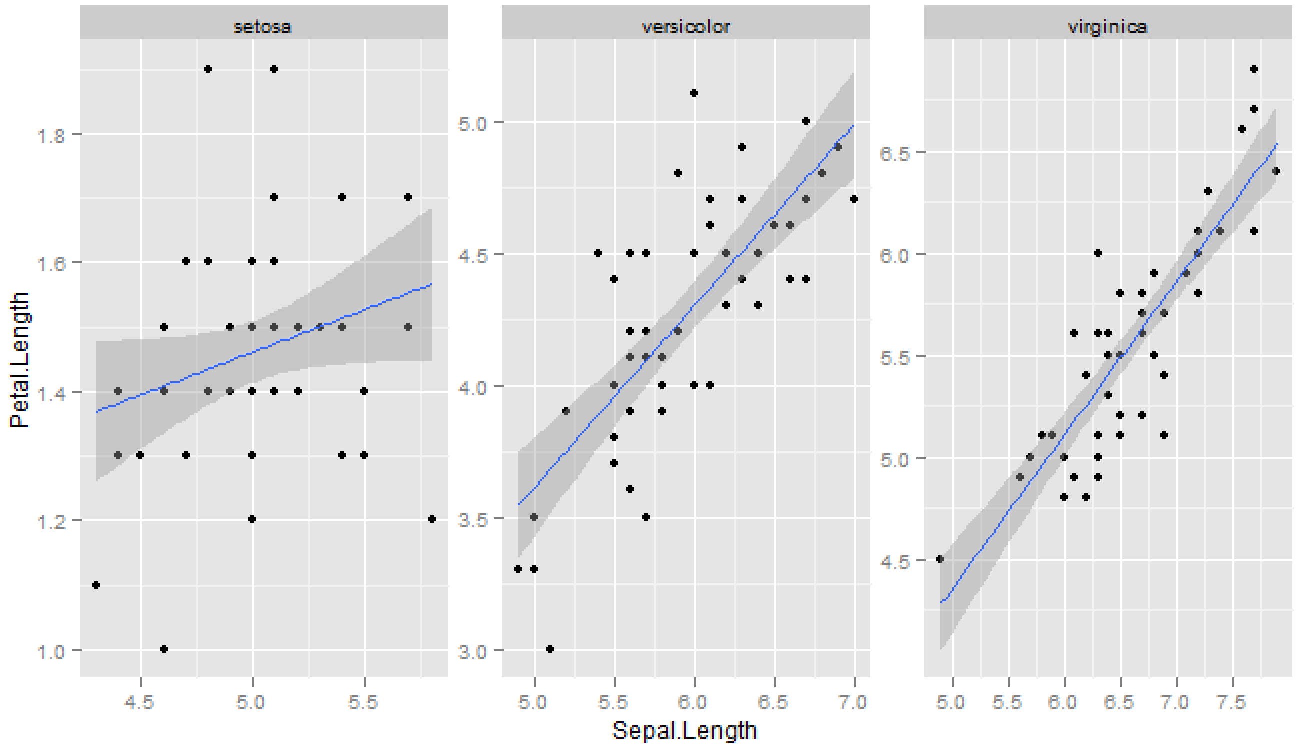

```
ggplot(iris, aes(x = Species, y = Sepal.Length)) +
  geom_boxplot(width=0.6) +
  geom_jitter(position=position_jitter(width=0.1),
  aes(col = Species))
```

DATA

GEOM

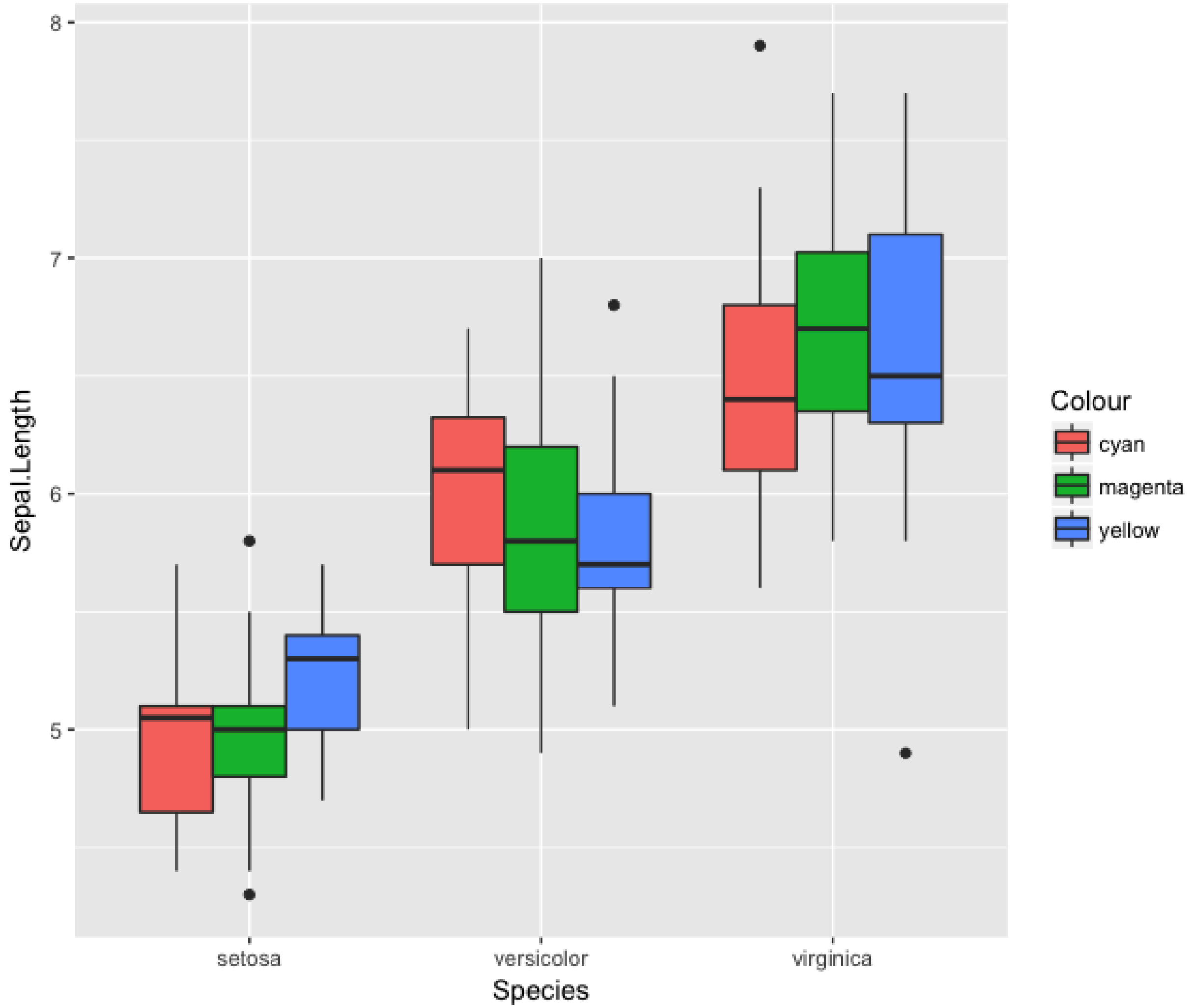
GEOM

5. Faceting



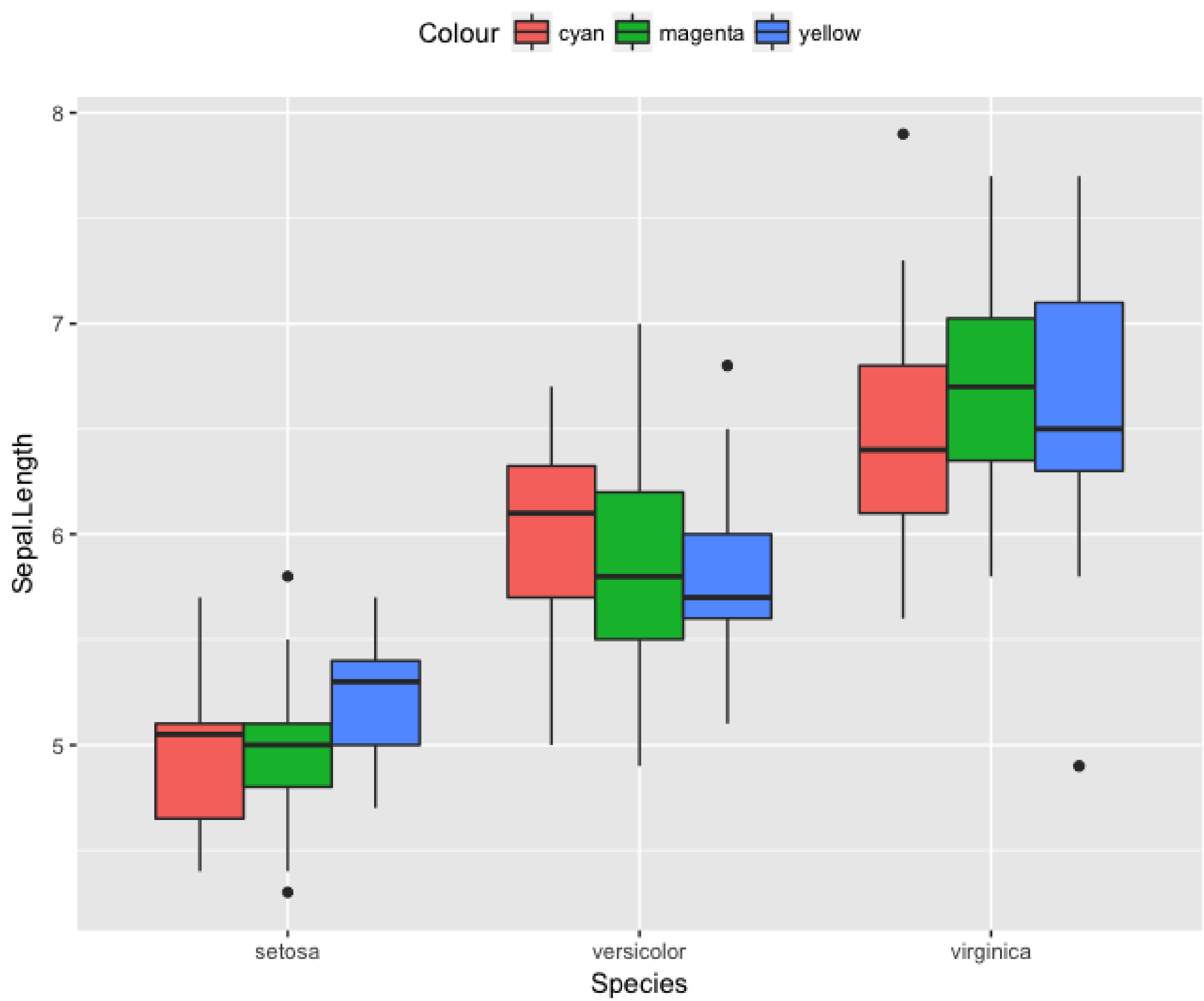

```
ggplot(iris, aes(x = Sepal.Length, y = Petal.Length)) +
  geom_point() +
  stat_smooth(method="lm") + facet_wrap(~ Species,
  scales = "free")
```

Divided by color




```
ggplot(iris2, aes(x = Species, y = Sepal.Length,
fill = Colour)) +   geom_boxplot()
```

6. Customisation




```
ggplot(iris2, aes(x = Species, y = Sepal.Length, fill =
 Colour)) +   geom_boxplot() + theme(legend.position='top')
```

- 更多內容參考GTW寫的《R ggplot2教學：基本概念與qplot函數》裡面ggplot繪圖架構那張圖，Science Craft上的文章《Introducing the Grammar of Graphics Plotting Concept》與Thomas Hopper寫的《A simple Introduction to the Graphing Philosophy of ggplot2》。

如果要練習需要資料的話，ggplot2這個套件也提供了10組資料，列表如下：

- diamonds：53,940顆原鑽的價格與特性
- economics：隨時間變化的美國經濟指標
- faithfuld：2d density estimate of Old Faithful data
- luv_colours：CIE 1976 (L*, u*, v*) 表色系，簡稱CIELUV。用L, u, v三個數值來描述一種顏色，另外還有附上顏色的名稱，總共列出了657種顏色
- midwest：中西部人口動態。參考《Midwest Demographic Analysis》
- mpg：從1999到2008，38種流行汽車的油耗資料
- msleep：83種哺乳動物的誰面資料
- presidential：從艾森豪到歐巴馬等11位美國總統的任期、所屬政黨
- seals：Vector field of seal movements
- txhousing：德州售屋資訊

HW6

- Please redo the plots using ggplot2 functions
  - Scatter plot for Image ID “770394” in HW3
  - Scatter plot for Image ID “770394” vs “236282” in HW3
  - 2D MDS & PCA plots in HW5